MILITARY SPECIFICATION

PERISCOPES, TANK: M28 SERIES

1. SCOPE

1.1 This specification covers monocular type periscopes used as sighting devices for machine guns mounted in the cupola of tanks or armored vehicles.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly and Inspection of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel: Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army, Frankford Arsenal

F8270601	Periscopes: M28, M28C and M28D
F10549640	Periscope: M28E1

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PACKAGING DATA SHEET

8270601	Packaging of Periscope, Tank: M28
8293432	Packaging of Periscope, Tank: M28C
8565458	Packaging of Periscope, Tank: M28D
10549640	Packaging of Periscope, Tank: M28E1

(Copies of specifications standards drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article testing.- Requirements for the submission of first article samples by the contractor for testing shall be as specified in the contract.

3.2 Fabrication.- The periscope shall be manufactured in accordance with the applicable drawing listed in Section 2 and drawings pertaining thereto.

3.3 General specifications.- The contractor shall be responsible for adherence to, and compliance with the following requirements of Specifications MIL-O-13830 and MIL-F-13926.

3.3.1 MIL-O-13830.

(a) Fixed eyepiece focus
(b) Cleanliness
(c) Illuminated reticle

3.3.2 MIL-F-13926.

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

3.4 Environmental.

3.4.1 Vibration.- The periscope shall be vibrated at an amplitude of 1/16 inch (1/8 inch maximum excursion) at a frequency of 30 cycles per second for a period of 5 minutes plus or minus 15 seconds and shall subsequently meet the requirements of 3.4.4.

3.4.2 Operating temperatures.- The periscope shall be exposed to the following temperatures:

(a) An ambient temperature of minus 40 degrees Fahrenheit for five hours, and shall then meet the requirements of 3.15.1.1 and 3.15.2 while at that temperature.

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(b) An ambient temperature of plus 150 degrees Fahrenheit for five hours, and shall then meet the requirements of 3.15.1.1 and 3.15.2 while at that temperature.

3.4.3 Storage temperatures.- The periscope shall be exposed to an ambient temperature of minus 80 degrees Fahrenheit for five hours, allowed to return gradually to room temperature and exposed to an ambient temperature of plus 160 degrees Fahrenheit for five hours. Upon re-establishment of thermal equilibrium at room temperature the periscope shall meet the requirements of 3.4.4.

3.4.4 Post environmental check.

3.4.4.1 After being vibrated, the periscope shall not be damaged, collimation shall not have changed by more than 0.25 mil from the vlaue established in 3.5 and the periscope shall meet all other requirements of this specification.

3.4.4.2 Cleanliness.- Subsequent to vibration the periscope shall meet the requirements of MIL-O-13830, and there shall be no more than 2 particles of dirt visible on the surfaces of the reticle and each particle shall not be in excess of an area equivalent to a size 10 dig. There shall be no dirt on other surfaces which interferes with observation or adversely affects image quality.

3.4.4.3 Each periscope that has been subjected to the requirements of 3.4.3 shall upon return to room temperature show no evidence of physical failure, such as cement separation, glass breakage or casting failure.

3.4.4.4 Each periscope that has been subjected to the requirements of 3.4.2 and 3.4.3 shall upon return to room temperature meet the following requirements:

(a) Collimation shall not have changed from the value established at Post Vibration (3.4.4.1) by more than 1.0 mil in any direction.

(b) The periscope shall meet all requirements of 3.6 to 3.15 inclusive.

3.5 Collimation.

3.5.1 Elevation.- The line of sight through the reticle boresight mark shall form an angle of 67 degrees 30 minutes plus or minus 1 mil with a line through the centers of the two bores of the elevation input arm assembly when the line of sight is parallel to the horizontal mounting surface of the instrument within 10 mils, and the elevation input arm adjustment is set at mid-point. The mid-point of the elevation boresight adjustment is defined as that setting which allows a

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minimum remaining adjustment of 1/16 inch in each direction.

3.5.2 Deflection.- The line of sight through the reticle boresight mark shall be parallel within 1 mil to a plane passing through the vertical geometric centerline of the sight body and the fixed center of the azimuth boresight eccentric when the line of sight is parallel to the horizontal mounting surface within 10 mils. The instrument shall be accepted with the eccentric oriented and locked in a position such that a plane thru the vertical geometric centerlines of the sight body and the large diameter of the eccentric is parallel to this established line of sight within 1 mil.

3.6 Parallax.- Parallax at the optical axis shall not exceed 0.50 mil when viewing a target between 25 meters and infinity.

3.7 Eyepiece focus.- The eyepiece focus shall be fixed between the limits of minus .75 and minus 1.0 diopter.

3.8 Resolution.- Resolving power on the optical axis shall be 30 seconds of arc or less.

3.9 Elevation tracking accuracy.- The angle between the line of sight through the reticle boresight mark and the line through the centers of the two bores of the elevation input arm (67 degrees 30 minutes nominal) shall remain constant within plus or minus 1.0 mil, excluding backlash, at any setting of the arm between 15 degrees depression and 30 degrees elevation, and within plus or minus 1.5 mils at any setting above 30 degrees elevation up to and including 60 degrees. The elevation or depression angle is defined as the angle between the line of sight and the horizontal mounting surface of the instrument.

3.10 Backlash.- Backlash between the optical line of sight and the elevation input arm shall not exceed 0.20 mil at any setting of the arm.

3.11 Elevation excursion limits.- Actuation of the elevation input arm shall cause an angular change in the line of sight from 15 degrees depression to 60 degrees elevation, a total excursion of 75 degrees. There shall be a full 48 degrees field of view with no evidence of cutoff at any setting.

3.12 Alinement.- With the periscope mounting surface perpendicular to a plumb line, the following conditions shall be met.

3.12.1 Plumb travel.- After establishing coincidence with a plumb line at 60 degrees elevation, the line of sight shall not deviate from the plumb line by more than 1.0 mil, nor shall the total lateral spread exceed 1.0 mil, when the line of sight is depressed through the total excursion of 75 degrees.

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3.12.2 Parallelism of reticle and image.- With the mounting surface of the periscope horizontal the 400 meter or 400 yard range line of the reticle shall be parallel to the image of a horizontal line within 24 minutes of arc. (Twenty-four minutes is equivalent to the angle subtended by the maximum thickness of a reticle line at one extremity of the 400 meter or 400 yard range line when the other extremity of the 400 meter or 400 yard range line is in coincidence with the image of the horizontal line.)

3.12.3 Image tilt.- The image of a vertical target shall be vertical within one degree. (One degree tilt is equivalent to the angle subtended by 14.3 mils at the edge of the field when measured over the full field of view.)

3.13 Reticle illumination.- When illuminated by an Instrument Light, M50 type or equal, the reticle markings shall appear clearly defined when observed in a darkened room.

3.14 Sealing.- The periscope shall show no evidence of leakage when subjected to an internal pressure of 5 psig for a period of five minutes. The interior atmosphere of the accepted periscope shall contain nitrogen having a dew-point at least as low as minus 25 degrees Fahrenheit at a pressure of between 0.1 and 0.25 psig.

3.15 Operability.

3.15.1 Torque.- The torque required to rotate the Linkage Arm Shaft B8270646 at room temperature (60 degrees to 90 degrees Fahrenheit) shall be as indicated on Drawing F8270601.

3.15.1.1 When measured at the temperatures specified in 3.4.2(a) and 3.4.2(b), torque shall not have changed from the value obtained at room temperature by more than five pound inches.

3.15.2 While exposed to the temperatures specified in 3.4.2(a) and 3.4.2(b), rotation of the input arm about its axis in either direction shall cause corresponding rotation of the head prism.

CAUTION: At no time shall the elevation input arm be allowed to snap (spring return) against the travel stop.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article testing.- When the contract specifies testing of preproduction or initial production samples, the following paragraphs shall apply as applicable.

4.2.1 Preproduction test.- Preproduction test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, two preproduction samples shall be submitted to the procuring agency for approval of design and construction features. Preproduction samples shall meet all inspection provisions of 4.3, 4.4, and 4.6 as well as any additional tests that may be specified in the contract.

4.2.2 Initial production test.- Initial production test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, samples shall be the first two production units and shall have been manufactured using the same materials, processes and procedures as will be used in regular production. Initial production testing shall be in accordance with 4.3, 4.4, and 4.6 and shall be witnessed by the Government representative for all examinations and tests. The Government reserves the right to conduct any or all examinations or tests as may be deemed necessary.

4.2.3 First article disapproval.- Failure of any one of a first article sample submitted shall be cause for disapproval by the Government and subsequent action as required by the contract clauses applicable to first article testing.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the Contracting Officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents (IEL), when provided with the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

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4.3.2.2 Final acceptance inspection.- Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I and II herein. Examination and tests for Packaging and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I and II shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

Use Level S-4 of Table I with
Sampling Plan Table II-A of Standard MIL-STD-105

		REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE			
MAJOR: AQL 2.5% Defective			
101.	Operability	3.4.2 & 3.15	4.6.5
MINOR: NONE			

NOTE: The above test shall be performed while the periscopes are thermally stabilized at temperatures of plus 150 degrees and minus 40 degrees Fahrenheit.

TABLE II - CLASSIFICATION OF DEFECTS

Use Level II of Table I with
Sampling Plan Table II-A of Standard MIL-STD-105

		REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE			
MAJOR: AQL 1.0%			
101.	Physical failure, post vibration	3.4.4.1	4.6.1.3.1
102.	Cleanliness, post vibration	3.4.4.2	4.6.1.3.2
103.	Collimation	3.5	4.6.2.1
104.	Parallax	3.6	4.6.2.2
105.	Backlash	3.10	4.6.2.3
106.	Elevation tracking accuracy	3.9	4.6.2.4
107.	Elevation excursion limits	3.11	4.6.2.5
108.	Plumb travel	3.12.1	4.6.2.6

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TABLE II - CLASSIFICATION OF DEFECTS (Cont'd)

		REQUIREMENT	TEST PROCEDURE
109.	Eyepiece focus	3.7	4.6.2.7.1
110.	Resolution	3.8	4.6.2.7.2
111.	Parallelism of reticle and image	3.12.2	4.6.2.7.3
112.	Image tilt	3.12.3	4.6.2.7.4
113.	Reticle illumination	3.13	4.6.3
114.	Sealing	3.14	4.6.4
115.	Torque	3.15.1	4.5.5.1

NOTE: The above tests shall be performed at standard ambient temperatures (+60° to +90°F).

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Control tests.

4.4.1 General.- One periscope shall be selected at random from each 25 produced, or from each months production, whichever occurs first. Control samples shall have successfully met all acceptance tests specified herein prior to conducting the following examinations and tests:

CONTROL TEST	REQUIREMENT	TEST PROCEDURE
Fabrication	3.2	Per applicable drawings and specifications
General specifications	3.3	Specifications MIL-O-13830 and MIL-F-13926
Storage temperatures	3.4.3	4.6.1.1

4.4.2 Operating temperatures.- Three periscopes shall be selected at random from each 50 produced or from each months production whichever occurs first, to determine operability in conformance with 3.4.2 and 4.6.5.

4.4.3 Control test failure.- Should the control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

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4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.3 Accuracy requirements for special testing devices.

4.5.3.1 Device to test vibration.- The complete periscope shall be vibrated on a vibration device which will produce a vertical motion at a constant frequency of 30 cycles per second plus or minus 5 cycles per second with an amplitude of 1/16 inch (1/8 inch plus or minus 1/64 inch maximum excursion). The vibration device shall be equipped with an adapter capable of holding the periscope rigidly to the vibration device in the normal operating position. The normal operating position is defined as follows: The periscope in the vertical position, the 2.9985 inch diameter bearing of the periscope body engaged in a 2.9990 plus .0005 minus .0000 inch diameter bore, both .502 inch diameter body locking pins engaged in a suitable camming device capable of locking the periscope against the horizontal mounting surface, the eccentric adjusting pin engaged in a suitable key slot, thus simulating the functional conditions of the applicable periscope mount, Drawing F8270627. A Periscope, Mount: M104A1, Drawing F8270627, may be used if modified to suit these requirements. In either case, the linkage arm shall be immobilized.

4.5.3.2 Device to test operating and storage temperatures.- The device shall consist of a chamber, the temperatures of which shall be in accordance with the temperatures specified in this specification. The temperature accuracies shall be within the tolerances specified under "Test facilities" in Specification MIL-F-13926.

4.5.3.3 Device to test collimation, elevation, azimuth, parallax, eyepiece focus, elevation tracking accuracy, backlash, elevation excursion limits and plumb travel.- A mandatory testing device shall be utilized in accordance with all requirements and specifications contained in and pertinent to drawings shown on IEL-F8270691.

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4.5.3.4 Device to test eyepiece focus, definition, parallelism of reticle and image and image tilt.- A mandatory device shall be utilized in accordance with all requirements and specifications contained in and pertinent to drawings shown on IEL-F8270601.

4.5.3.5 Device to test torque.- The device shall consist of a torque wrench type tester which will permit reading in both the clockwise and counterclockwise directions with a minimum range of from zero to 15 pound-inches; the accuracy of which shall be within plus or minus 1/4 pound-inch throughout the entire zone of excursion. The device shall be equipped with a suitable adapter to engage the input arm in order to accommodate the required test.

4.5.3.6 Device to test sealing.- The device to test sealing shall be capable of charging the interior of the periscope to an internal pressure of at least 5 psig of nitrogen having a dew-point of at least as low as 25 degrees Fahrenheit. The device shall be equipped with a pressure gage capable of recording the internal pressures up to at least 5 psig, sub-divided into increments of .1 psig. The accuracy of the pressure gage shall be within 1 percent at any point on the scale.

4.6 Test methods and procedures.

4.6.1 Environmental tests.

4.6.1.1 Storage temperatures.- This test shall be conducted in accordance with 3.4.3, utilizing the conditions specified in 4.5.2.3.2. Upon completion of the specified exposure, the periscope shall be visually inspected for damage and then subjected to all the tests specified in this specification.

4.6.1.2 Operating temperatures.- The tests conducted at plus 150 degrees and minus 40 degrees Fahrenheit shall be performed in accordance with 4.5.2.3.2 and 4.6.5 to determine conformance with 3.4.2.

4.6.1.3 Vibration.- Each periscope in the sample shall be mounted on a special testing device (4.5.2.3.1) and vibrated in accordance with the requirements of 3.4.3.

4.6.1.3.1 Post vibration.- Subsequent to vibration, the periscope shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine compliance with 3.4.4. A visual and tactile examination shall be made for missing or damaged parts. Minimum visual examination shall include all springs, screws, clamps, sealing, etc. A tactile examination shall be conducted to determine looseness or malfunction of above items or assembled members, to include all moving parts such as arms, elevation assemblies, etc. An audible examination shall be made by holding the periscope close to the ear and vigorously shaking it with both hands while listening for looseness of internal optical and mechanical parts. Any one of the above defects shall be cause to reject the periscope as defective, unless separate defects-per-hundred type sampling plans are established

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and approved by the Government as part of the suppliers inspection system. Periscopes which have passed the above examinations shall be subjected to the tests of 4.6.2 to 4.6.5.3 inclusive to determine compliance with 3.4.4.1.

4.6.1.3.2 Cleanliness, post vibration.- This test shall be conducted in accordance with Specification MIL-O-13830. There shall be no visible evidence of moisture on the internal surfaces of the periscope. Dirt particles shall not exceed the tolerance specified in 3.4.3.1.

4.6.2 Performance tests.

4.6.2.1 Collimation.- Position the periscope in test fixture F8289541 as specified on Drawing B8262159. Sight through the eyepiece of the periscope. The reflected image of the reticle borecross mark of the periscope shall be within the inner square of the fixture reticle to meet the specified tolerance in 3.5 for deflection and elevation.

4.6.2.2 Parallax.- Parallax shall be tested with the periscope still mounted in the test fixture. The center of the borecross on the periscope's reticle shall be superimposed on the dot in the center of the circle of the fixture reticle. The observer will sight through the eyepiece of the periscope. Parallax will be recognized as any apparent movement of the borecross in relation to the fixture reticle circle when the observer's head is moved side to side and up and down. Parallax shall not exceed the specified tolerance in 3.6.

4.6.2.3 Backlash.- Elevate the periscope line of sight, with backlash excluded, to zero elevation as indicated on the test fixture. Superimpose the fixture reticle dot on the boresight mark. Record the test fixture reticle micrometer setting. Continue elevation of the line of sight to plus 30 degrees; superimpose the reticle imates and record micrometer settings. Continue elevation to plus 60 degrees; superimpose the reticle images and record micrometer settings. Continue in elevation past the 60 degree elevation mark; then depress the line of sight with the backlash included. When the line of sight is depressed to 60 degrees, 30 degrees and zero degrees of elevation, and the fixture reticle is reset to the recorded readings for those positions, the boresight mark shall not be separated from the fixture reticle dot by more than the distance between the fixture reticle dot and the inner limits of the short horizontal line intersecting the circle. The backlash shall be measured for the excursion from zero degrees to minus 15 degrees by using the same method to determine conformance with 3.10.

4.6.2.4 Elevation tracking accuracy.- Elevate the periscope line of sight, with backlash excluded, to the zero elevation position as indicated on the fixture. At this position, the images of the fixture reticle dot and the borecross of the periscope reticle shall be superimposed. Depress the line of sight beyond the minus 15 degree position to eliminate backlash for the elevation excursion. At the minus 15 degree, and at any setting of the elevation input arm between minus 15 degrees, and plus 30 degrees elevation, the center of the borecross

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shall remain within the inner square of the fixture reticle. At any setting of the elevation input arm between plus 30 degrees and plus 60 degrees elevation, the center of the borecross shall not move beyond the limits of the rectangular reticle pattern of the fixture to determine conformance with 3.9.

4.6.2.5 Elevation excursion limits.- This test may be conducted concurrent with the elevation tracking accuracy test. Actuation of the elevation input arm from minus 15 degrees to the plus 60 degree position, as measured on the fixed counter, shall cause a corresponding change in the line of sight in compliance with 3.11.

4.6.2.6 Plumb travel.- With the periscope positioned in the test fixture, the periscope line of sight shall be elevated to plus 60 degrees and the fixture reticle dot superimposed on the center of the borecross of the periscope reticle. The line of sight shall be depressed to minus 15 degrees, a total of 75 degrees excursion. During this excursion, the fixture reticle dot shall be depressed so that the dot and center of the borecross remain in the same plane for the total excursion within the specified tolerance in 3.12.1.

4.6.2.7 Orientation tests.- The following tests shall be conducted with the periscope positioned in test fixture F8615059 as specified on Drawing B8618035. The target for these tests shall be as specified on Drawing D8272086. The test fixture and target shall be set up as specified on Drawing B8618034.

4.6.2.7.1 Eyepiece focus.- Adjust the eyepiece of the dioptometer for best focus on the dioptometer reticle, with the diopter scale set at zero. Sight through the dioptometer while positioned to the rear of the eyepiece of the periscope. Focus the dioptometer to bring the reticle of the periscope into sharpest focus. The reading taken from the diopter scale shall be within the tolerance as specified in 3.7.

4.6.2.7.2 Resolution.- Adjust the eyepiece of the dioptometer for best focus on its reticle. While viewing the parallel bar test pattern through the eyepiece of the dioptometer, adjust the objective end of the dioptometer for best focus. The parallel bar test pattern shall be clearly defined in each of the four meridians with the specified tolerance. Illumination of the test target shall be a minimum of ten millilamberts. At least three regions of the clear aperture of the periscope, mutually separated by 1-1/2 inches or more, shall be inspected for the resolution requirements of 3.8.

4.6.2.7.3 Parallelism of reticle and image.- While viewing the test target through the eyepiece of the dioptometer, the 400 meter or 400 yard range line of the periscope reticle shall be parallel to the horizontal target line. One end of the 400 meter or 400 yard range line shall be in coincidence with the horizontal target line and the other end of the range line shall not exceed the specified tolerance in 3.12.2. The dioptometer is only used for viewing during this test.

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4.6.2.7.4 Image tilt.- While viewing the test target through the eyepiece of the dioptometer, the vertical line of the target image shall be parallel to the vertical line of the dioptometer reticle line within the tolerance on one end of the target in accordance with 3.12.3, when the line images intersect at the other end.

4.6.3 Reticle illumination.- The reticle markings shall be visually checked in a darkened room and shall meet the specified requirement of 3.13. The reticle shall be illuminated with an Instrument Light M50, or in the absence of an instrument light, a device consisting of two 1-1/2 volt batteries with a .35 to .5 amp bulb may be used.

4.6.4 Sealing.- The periscope shall be pressurized to the specified requirement to determine leakage in compliance with 3.14. The interior pressure shall not decrease due to leakage when checked by suitable measuring equipment. Upon satisfactory completion of the sealing test, the periscope shall be flushed with dry nitrogen. The flushing shall continue until the gas bled from the periscope has a dew-point at least as low as the pressure specified. The periscope shall then be sealed with an internal pressure as specified. This test shall be performed using a special testing device.

4.6.5 Operability.

4.6.5.1 Torque.- This test shall be conducted using a special testing device. The device shall be affixed to the linkage arm shaft. The torque required to rotate the linkage arm shaft at room temperature shall be measured and recorded. It shall meet the specified requirements of 3.15.1.

4.6.5.2 The periscope shall then be exposed to the extreme temperatures as specified in 3.4.2. At these temperatures, the torque shall again be measured and recorded. It shall not deviate from the previously recorded torque more than the specified tolerance in 3.15.1.1.

4.6.5.3 During exposure at the specified ambient temperatures, rotation of the input arm about its axis in either direction shall be visually checked to ascertain of this rotation causes corresponding rotation of the head prism in accordance with 3.15.2.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Specification MIL-P-14232 and the applicable packaging data sheet listed in Section 2. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The periscope M28 and M28C are utilized as

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sighting devices for a 50 caliber machine gun mounted in the cupola of a tank or armored vehicle. The M28 Periscope has a ballistic reticle calibrated in yards, the M28C has a ballistics reticle calibrated in meters. The periscope M28D is the primary sighting instrument for the M113 Flame Thrower Vehicle for use with the main flame gun and employs a reticle inscribed with ballistics data for both the flame gun and the machine gun. The M28E1 Periscope is used on the Armored Assault Vehicle XM729.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of packaging and packing.
(c) Applicable Packaging Data Sheet number (see 5.1).
(d) Applicable stock number.

Custodian:

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Preparing activity:

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Project No. 1240-A571Y

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